

Customer Segmentation Report

Using RFM Analysis and Clustering

Project Overview

This project segments customers based on their purchasing behavior using Recency, Frequency, and Monetary (RFM) features. The goal is to identify customer groups that can be targeted with tailored marketing strategies.

Dataset Summary

Source: UCI Online Retail Dataset

Period: Dec 2010 - Dec 2011

Records: ~500,000 transactions

Customers analyzed: ~4,300 (after cleaning)

RFM Features

- Recency: Days since last purchase
- Frequency: Number of purchases
- Monetary: Total amount spent

The RFM data was log-transformed and scaled before applying clustering.

Clustering Techniques

- KMeans (Elbow Method to select k)
- Hierarchical Clustering (Ward linkage, Dendrogram)
- DBSCAN (Density-based clustering with outlier detection)

PCA was used to visualize clusters in 2D.

Cluster Insights

Example interpretation of customer segments:

Cluster 0 - Loyal Big Spenders

Cluster 1 - At-Risk / Lost Customers

Cluster 2 - Occasional Buyers

Cluster 3 - New Big Spenders

Business Recommendations

- Target Cluster 0 with loyalty programs and VIP perks
- Re-engage Cluster 1 with special discounts or reminders
- Upsell to Cluster 2 based on browsing/purchase behavior
- Welcome and nurture Cluster 3 for retention and conversion

Next Steps

- Integrate with CRM for live customer segmentation
- Automate re-segmentation over time
- Track segment migration to improve targeting